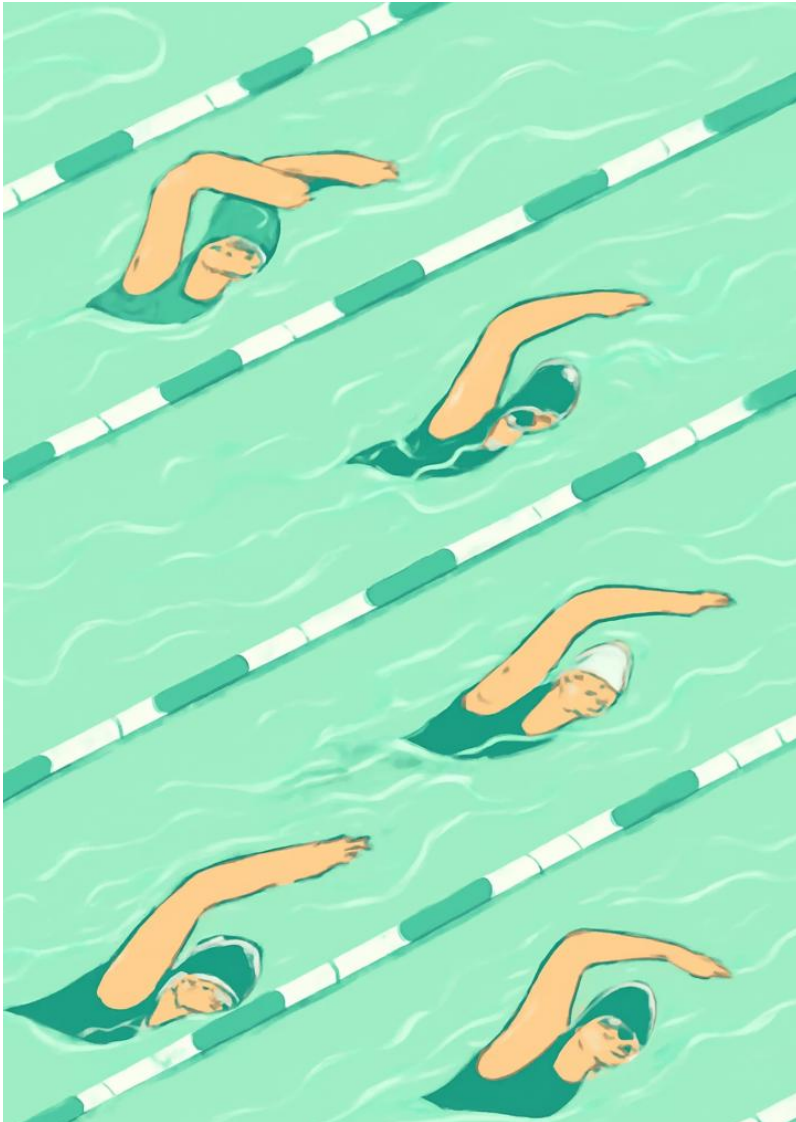




Linear Motion

Motion in a Straight Line – the simplest kind of motion, where an object moves along a single straight path.

What is Linear Motion?

When an object moves in a straight line, its motion is called **linear motion**. A falling ball, a swimmer in a lane, a car on a highway – all examples of this.

Position & Reference Point

To describe motion, we first fix a **reference point**. The distance and direction of an object from that point – at any instant – gives its **position**.

Motion vs. Rest

If position changes with time → **motion**. If position stays the same → **rest**. Both are always defined relative to the reference point.

Examples of Linear Motion

Motion All Around Us



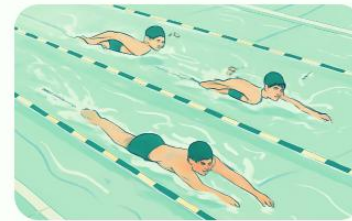
Falling Ball

A ball dropped vertically falls in a perfectly straight line – classic linear motion.



Car on Highway

A car travelling along a straight stretch of road moves in one direction without turning.



Swimming Race

Swimmers in straight lanes move from one end to the other – pure linear motion.

Examples of Linear Motion

Motion All Around Us



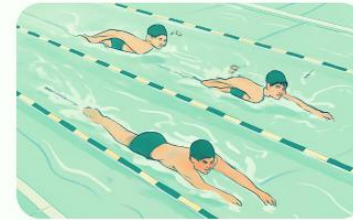
Falling Ball

A ball dropped vertically falls in a perfectly straight line – classic linear motion.



Car on Highway

A car travelling along a straight stretch of road moves in one direction without turning.

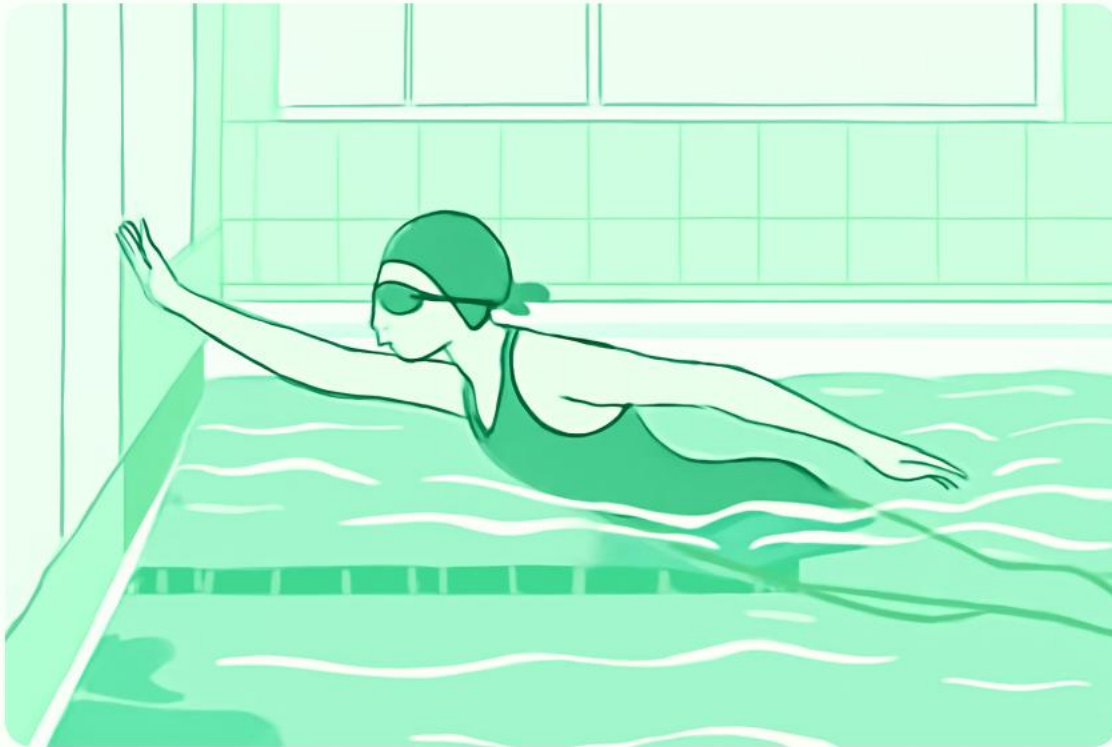


Swimming Race

Swimmers in straight lanes move from one end to the other – pure linear motion.

Speed vs. Velocity

Average Speed & Average Velocity



Sarang's Swim — 50 Seconds

Sarang swims the length of a 25 m pool and returns to his starting point. His total journey takes exactly 50 seconds.

Total Distance - $25 + 25 = 50 \text{ m}$

Displacement - Start $\rightarrow$ Start - 0 m

Average Speed - $50 \text{ m} \div 50 \text{ s} = 1 \text{ m/s}$

Average Velocity - $0 \text{ m} \div 50 \text{ s} = 0 \text{ m/s}$

i Average speed can be non-zero even when average velocity is zero – because he returns to his starting point!

Speed $\neq$ Velocity

Two quantities that sound similar – but measure very different things.

Average Speed

Total distance travelled divided by the time taken. It is always **positive** and has no direction.

Formula: $\text{Speed} = \text{Total Distance} / \text{Time}$

Average Velocity

Total displacement divided by the time taken. It **can be zero or negative** depending on direction.

Formula: $\text{Velocity} = \text{Displacement} / \text{Time}$

❏ Displacement has both magnitude and direction – making velocity a **vector** quantity, while speed is a **scalar**.

Average Acceleration

The Bus Problem

A bus travelling at 36 km/h (10 m/s) speeds up, then brakes to a stop. Two different accelerations occur.

① Accelerator Pressed

$$u = 10 \text{ m/s} \rightarrow v = 15 \text{ m/s}, t = 10 \text{ s}$$

$$a = (15 - 10) / 10 = +0.5 \text{ m/s}^2$$

Positive – acceleration acts *in the direction of motion*.

② Brakes Applied

$$u = 15 \text{ m/s} \rightarrow v = 0 \text{ m/s}, t = 5 \text{ s}$$

$$a = (0 - 15) / 5 = -3 \text{ m/s}^2$$

Negative – acceleration acts *opposite to the direction of motion*.

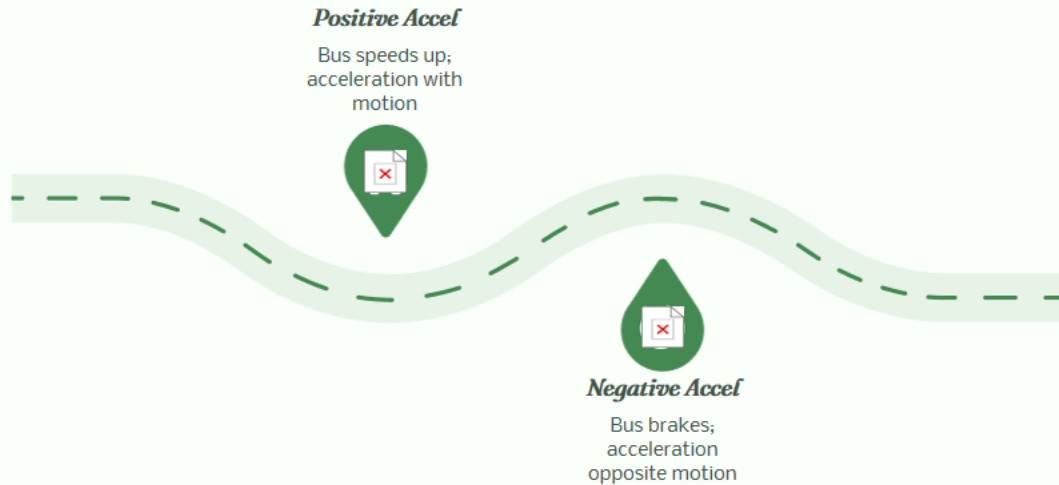


Understanding Acceleration

The Formula

$$a = \frac{v - u}{t}$$

Where **u** = initial velocity, **v** = final velocity, **t** = time interval.



Positive Acceleration

Speed is **increasing**.
Force acts in the same direction as motion.

Negative Acceleration

Speed is **decreasing** (deceleration). Force acts against motion.



Uniform Circular Motion

Moving in Circles

When an object moves in a circular path at **constant speed**, its motion is called **uniform circular motion**. The speed stays the same – but the direction of velocity changes continuously.

Speed

Constant at every point on the circle.

Direction

Always along the **tangent** to the circle at that point – continuously changing.

Acceleration

Present – because velocity's direction changes, even if speed is constant.

Formula

$v = 2\pi R / T$ where R = radius, T = time period.

The Marble & Ring Activity



What Happens?

Place a ring flat on a smooth surface and roll a marble along its inner edge. The marble follows a circular path.

Now **lift the ring**. The marble immediately moves in a **straight line** – in the direction it was heading at that exact instant.

- ✔ This proves that velocity in circular motion points along the **tangent**. Without the ring to redirect it, the marble travels in a straight line – Newton's First Law at work!

Chapter Summary

Describing Motion — Key Takeaways

Concept	Key Idea	Formula
Linear Motion	Motion along a straight line; position defined from a reference point	–
Distance vs. Displacement	Distance = total path; Displacement = net change with direction	Displacement = Final – Initial position
Average Speed	Total distance ÷ time; always positive; no direction	$s = d / t$
Average Velocity	Displacement ÷ time; can be zero or negative	$v = \Delta x / t$
Average Acceleration	Change in velocity ÷ time; negative means deceleration	$a = (v - u) / t$
Uniform Circular Motion	Constant speed, changing direction; always accelerating	$v = 2\pi R / T$

- ❏ Remember: **Scalars** have magnitude only (distance, speed). **Vectors** have magnitude AND direction (displacement, velocity, acceleration).